

Revealing Hidden Ripples with PCA Residuals

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This document demonstrates a common signal-processing insight: a tiny, high-frequency ripple added to a large sine wave is almost invisible in the raw waveform, but becomes obvious once we remove the shared structure using PCA residuals.

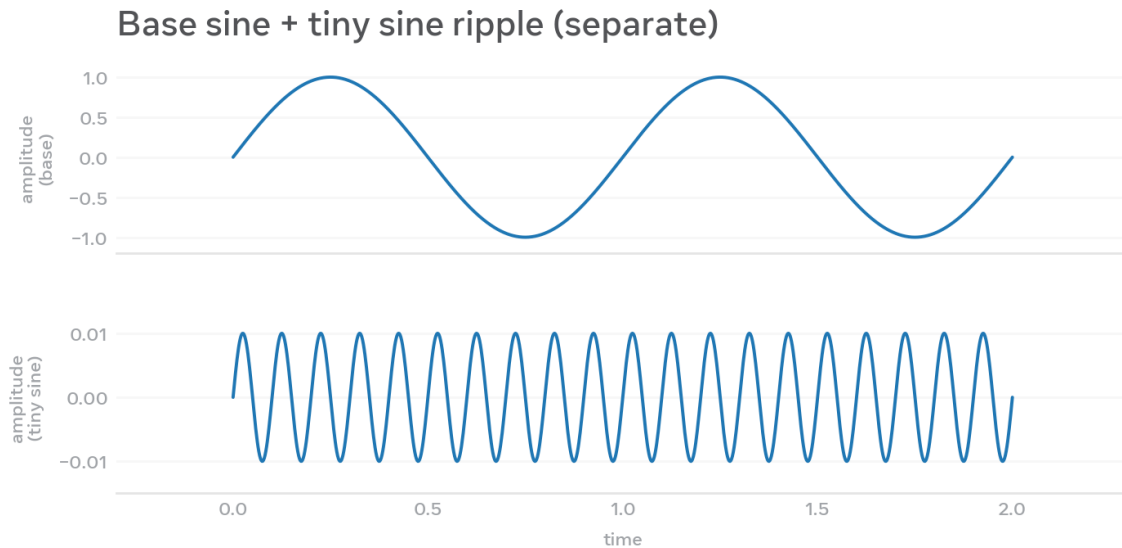
1. Concept

We start with a base sine wave at frequency $f=1$ Hz (amplitude 1.0) over 2 seconds (1000 samples). To this we add a second signal that is only 1% as large (amplitude 0.01) but at 10x the frequency (10 Hz). Four variants of the tiny signal are tested: sine, square, triangle, and sawtooth. Because the ripple is 100x smaller, the combined waveform looks almost identical to the pure sine.

2. Preview – Base and Ripple Separate

The first four images show the two components separately in a double graph. Top: the large base sine. Bottom: the tiny 1% ripple (y-axis ± 0.015). This establishes what we are about to hide inside the sine.

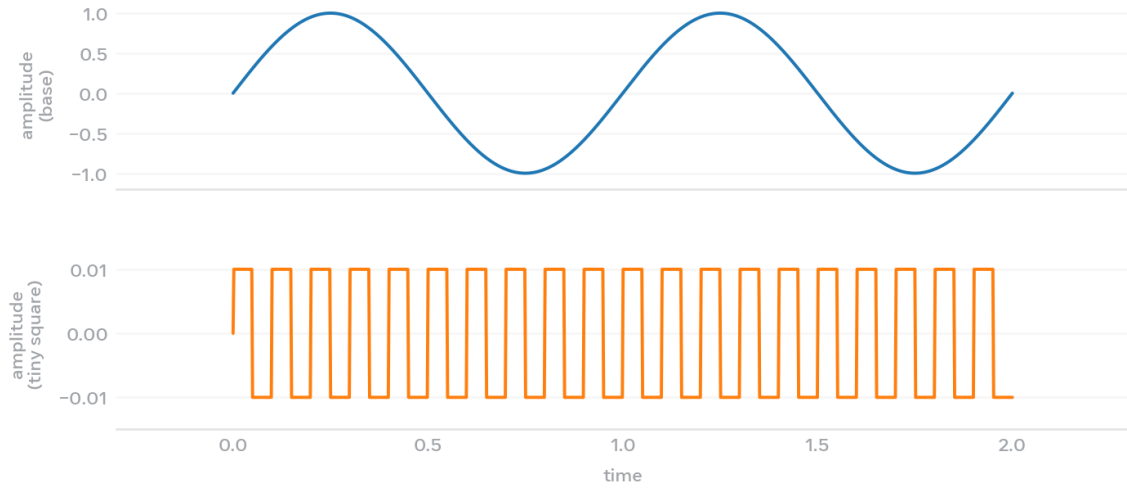
Preview – Sine Ripple: Base sine (top) and $0.01 * \text{sine at } 10\text{x freq}$ (bottom).



PREVIEW_SINE BASE_AND_RIPPLE_SEPARATE [EDITABLE LABEL]

Preview – Square Ripple: Base sine (top) and $0.01 * \text{square at } 10\text{x freq}$ (bottom).

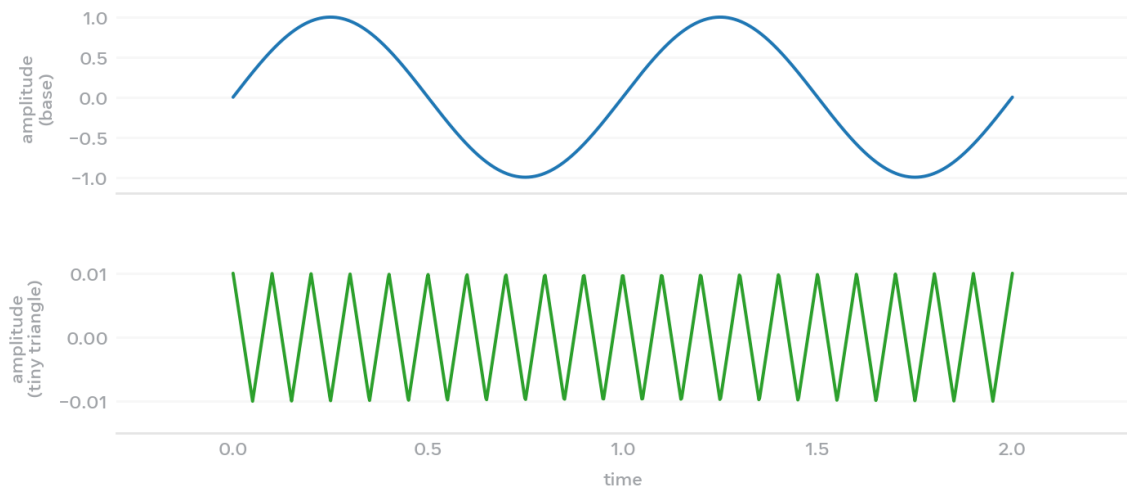
Base sine + tiny square ripple (separate)



PREVIEW_SQUARE BASE_AND_RIPPLE_SEPARATE [EDITABLE LABEL]

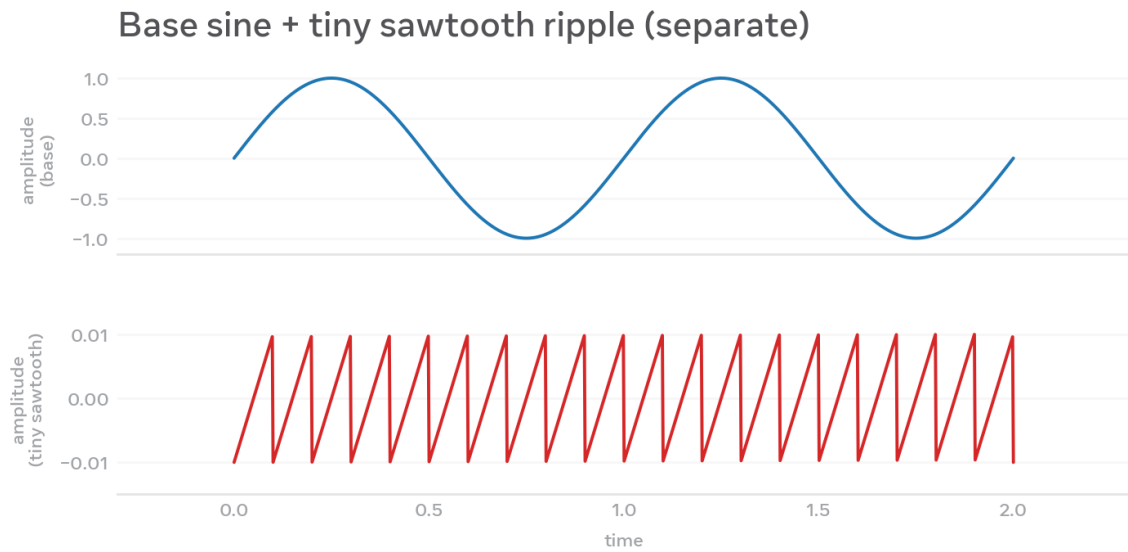
Preview – Triangle Ripple: Base sine (top) and $0.01 \times$ triangle at 10x freq (bottom).

Base sine + tiny triangle ripple (separate)



PREVIEW_TRIANGLE BASE_AND_RIPPLE_SEPARATE [EDITABLE LABEL]

Preview – Sawtooth Ripple: Base sine (top) and $0.01 \times$ sawtooth at 10x freq (bottom).

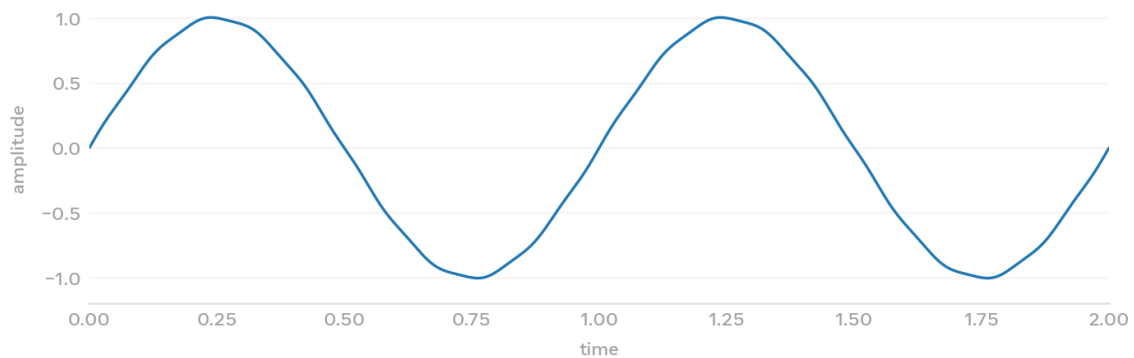


PREVIEW_SAWTOOTH BASE_AND_RIPPLE_SEPARATE [EDITABLE LABEL]

3. Combined Signal – Ripple Almost Invisible

Next four images show sine + ripple. At full scale ($y \pm 1.2$), the 1% high-frequency variation is nearly indistinguishable. This mimics real-world cases where a dominant common mode masks small but important differences.

Original Combined – sine + $0.01 \cdot \text{sine}(10x)$. The ripple is present but visually lost.



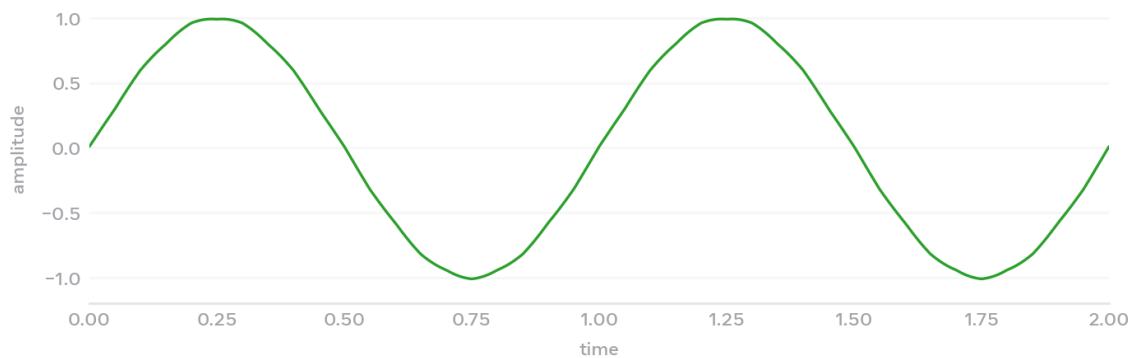
ORIGINAL_SINE [EDITABLE LABEL]

Original Combined – sine + $0.01 \cdot \text{square}(10x)$. The ripple is present but visually lost.



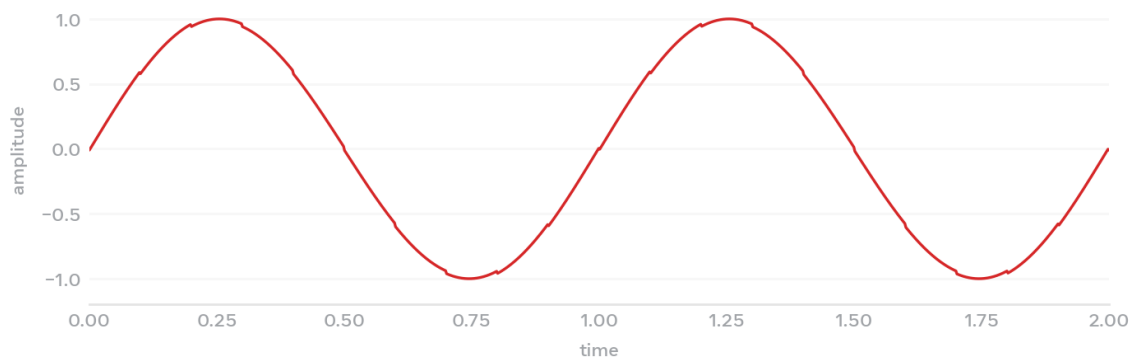
ORIGINAL_SQUARE [EDITABLE LABEL]

Original Combined – sine + 0.01*triangle (10x). The ripple is present but visually lost.



ORIGINAL_TRIANGLE [EDITABLE LABEL]

Original Combined – sine + 0.01*sawtooth (10x). The ripple is present but visually lost.



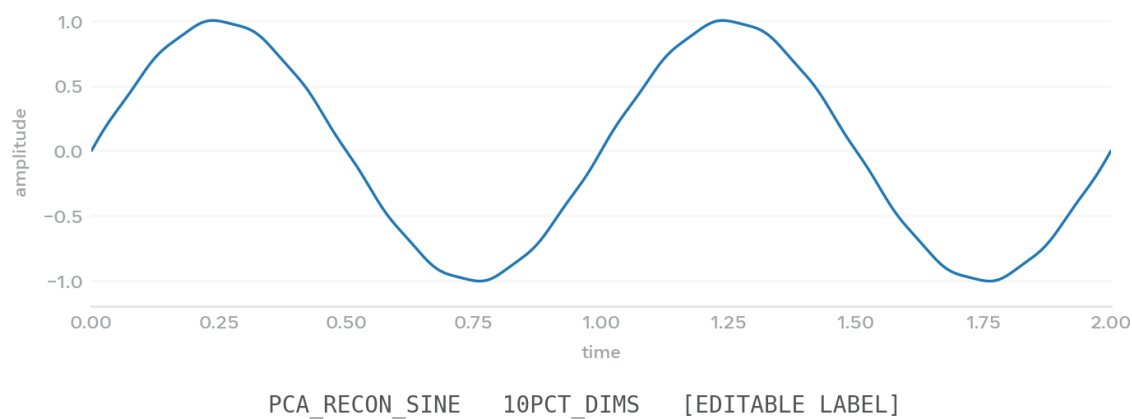
ORIGINAL_SAWTOOTH [EDITABLE LABEL]

4. PCA Compression to 10% Dimensions

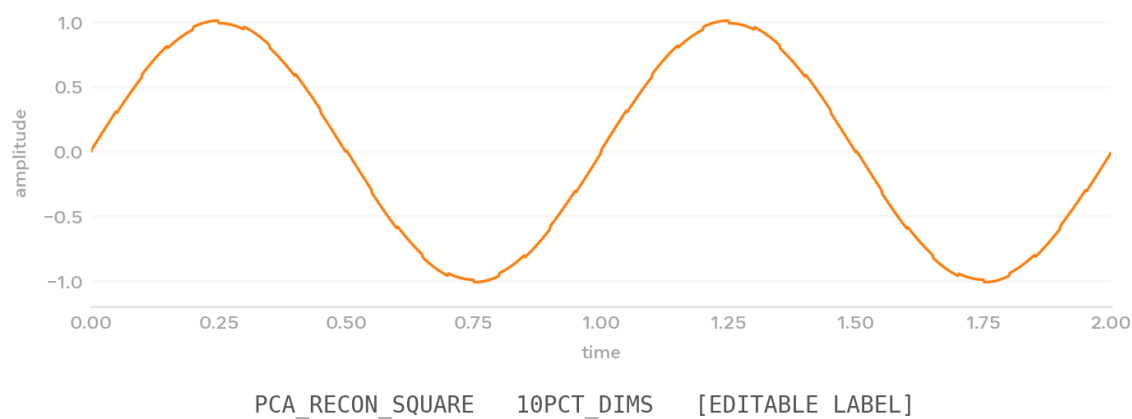
We stack the four combined signals into a 4 x 1000 matrix and apply PCA. We keep 4 components (far below the 10% target of 100 dimensions for a 1000-point vector). Because the

four waveforms share the same large sine, their intrinsic rank is low. The PCA reconstruction is therefore almost perfect – it recreates the original sine+ripple graphs with very little error, showing that dimensionality reduction preserves the dominant structure.

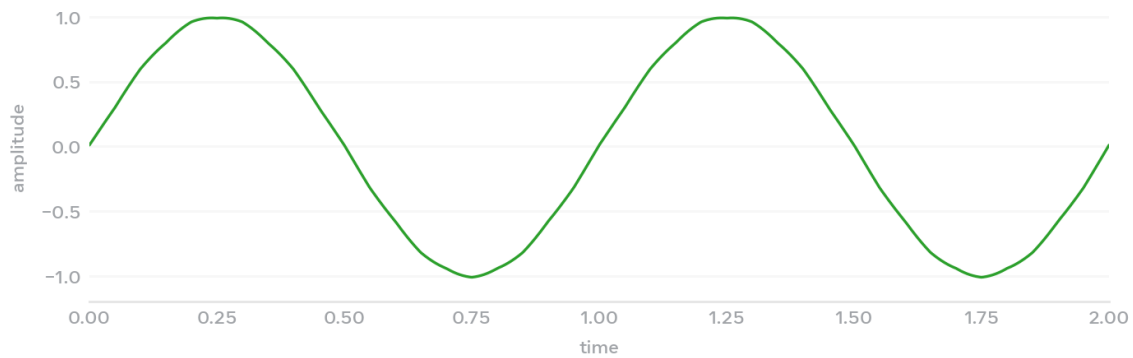
PCA Reconstruction – sine: Inverse-transformed from PCA space. Good recreation of original.



PCA Reconstruction – square: Inverse-transformed from PCA space. Good recreation of original.

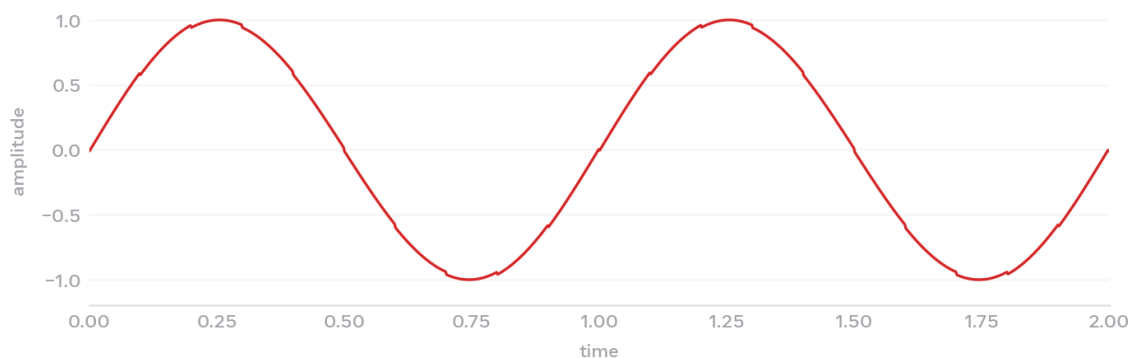


PCA Reconstruction – triangle: Inverse-transformed from PCA space. Good recreation of original.



PCA_RECON_TRIANGLE 10PCT_DIMS [EDITABLE LABEL]

PCA Reconstruction – sawtooth: Inverse-transformed from PCA space. Good recreation of original.

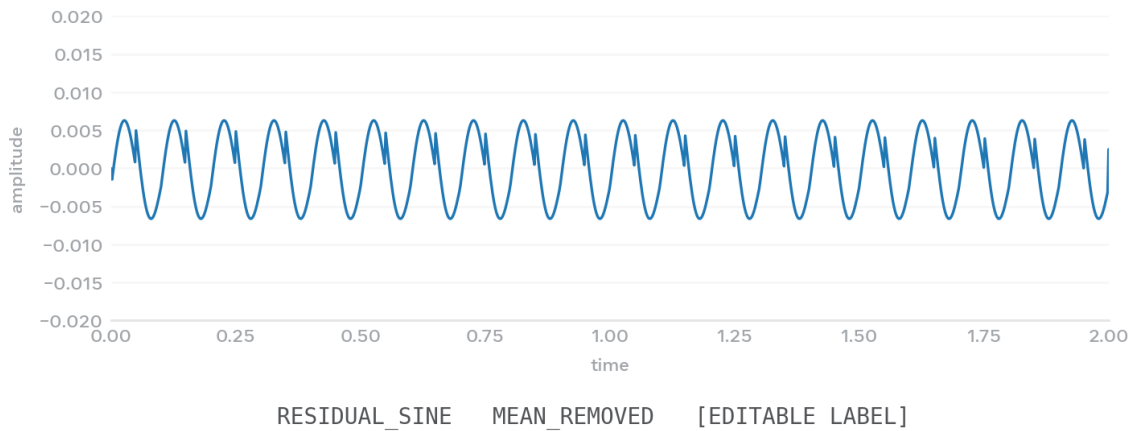


PCA_RECON_SAWTOOTH 10PCT_DIMS [EDITABLE LABEL]

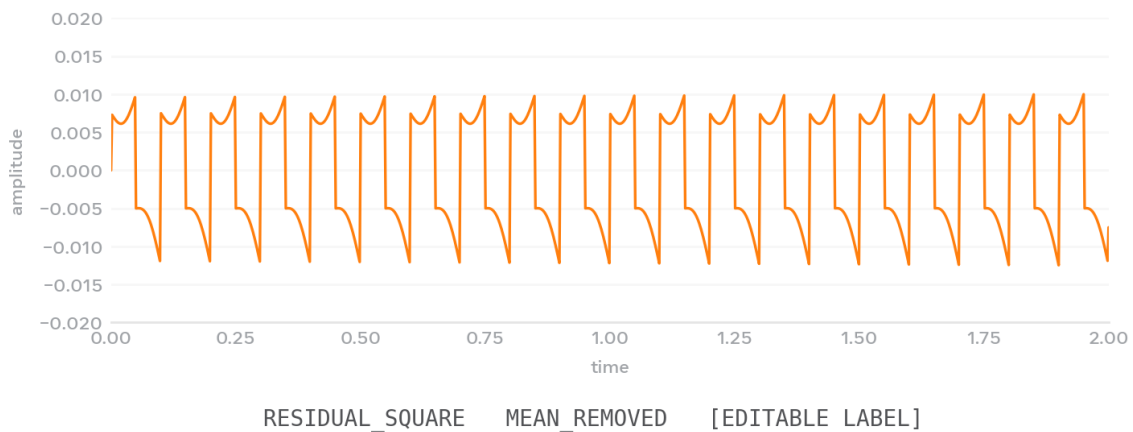
5. Residual PCA – Removing the Average Reveals the Ripple

This is the key step. We compute the mean of the four PCA embeddings (mean_Z) and subtract it from each individual embedding: $Z_{\text{res}} = Z_i - \text{mean_Z}$. When we reconstruct only from Z_{res} (Z_{res} @ components, without adding the global mean back), we are reconstructing $(X_i - \text{mean}(X))$ in the PCA subspace. Since $\text{mean}(X)$ is essentially the common large sine wave, it is removed. The y-axis is now ± 0.02 , a 60x zoom. The tiny ripple – sine, square, triangle, sawtooth at 10x – is now clearly visible and correctly shaped. This shows how mean-centered PCA residuals can be used to isolate and visualize small systematic differences hidden under a large common mode.

Residual PCA – sine: Mean removed. Common sine gone, 1% sine ripple now clearly visible.



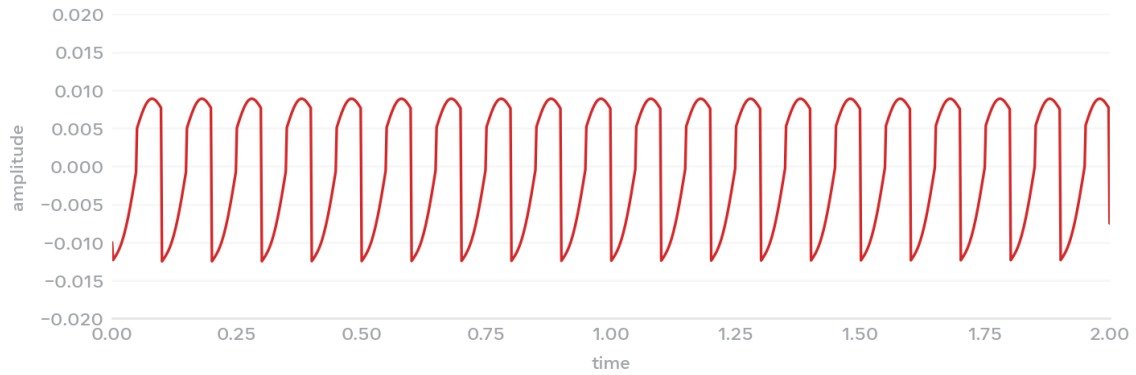
Residual PCA – square: Mean removed. Common sine gone, 1% square ripple now clearly visible.



Residual PCA – triangle: Mean removed. Common sine gone, 1% triangle ripple now clearly visible.



Residual PCA – sawtooth: Mean removed. Common sine gone, 1% sawtooth ripple now clearly visible.



RESIDUAL_SAWTOOTH MEAN_REMOVED [EDITABLE LABEL]

Summary

1) Small high-frequency signals (1%, 10x) are invisible on top of a large sine. 2) PCA to 10% of dimensions (100 $\rightarrow$ 4 effective) still recreates them well. 3) Subtracting the mean PCA embedding and reconstructing the residual removes the shared sine and exposes the hidden waveform. This technique generalizes to any set of signals sharing a dominant common mode.